

Adaptive Preference Discovery Agent



2026. 09. 13.

Subject	Human Computer Interaction
Team Name	Handsome Team
Team Leader	Jaehyun Kim
Team Member	Kyeongho Kim, Jonghyun Noh, Seyong Park, Junsang Lee

1

Project Topic

A. Topic

This project proposes an AI fashion recommendation agent that distinguishes between a user's relatively persistent style preferences and transient needs that arise from their current situation and continuously reconciles the two over the course of a conversation. On top of a baseline style profile that the user can review and revise, the system incorporates session-level context such as schedule, activity, location, and desired impression. The agent then determines which preferences are currently uncertain and presents either a dialogue or outfit cards depending on whether that preference is more easily expressed in language or more suited to visual comparison. By iteratively incorporating the user's natural language responses and image-based like/dislike selections, the system refines the current intent, and once it determines that sufficient information has been gathered, it transitions to a final recommendation. Through this, the project explores how user modeling that separates baseline preferences from transient context, combined with an interaction that adaptively integrates dialogue and image selection, affects the user's ability to express intent, sense of control, and overall recommendation experience.

B. Hypothesis

- **H1: Effect of separating and reconciling baseline preferences and context**

Compared to an approach that adjusts the two types of information without distinguishing between them, an approach in which users separately review and revise their baseline preferences and current context increases their sense of control and helps them make choices that satisfy both authenticity (feeling "like themselves") and situational appropriateness.

- **H2: Effect of adaptive interaction**

Compared to confirming all preferences through text-based dialogue alone, or presenting images in a fixed manner, an approach that adaptively switches between dialogue and image-based selection depending on the type of preference and the system's level of uncertainty reduces the burden of expressing one's intent and increases the degree to which users feel the AI has understood their intent.

2-1

Persona #1

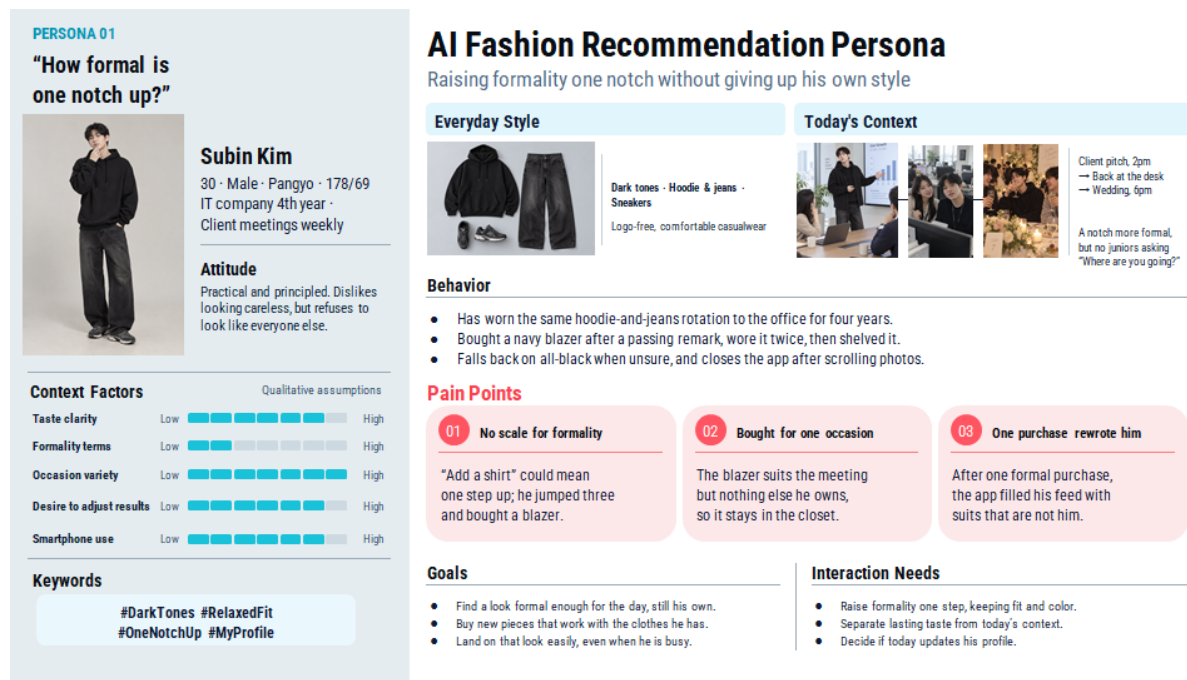


Figure1. Summary of Persona1

Description

Dark-toned clothing makes up 90% of his closet; he doesn't prefer highly saturated colors. Hoodies and jeans have been his go-to for 4 years running, and he avoids buying anything with a body-hugging fit because it feels uncomfortable. Clothes with logos, patterns, or decorative details bother him all day if he wears them. The only formal outfit he owns is a black set he bought last month for a wedding. He's aware of trends but doesn't follow them. He mainly shops on Musinsa.

2-2

Scenario #1

A. Problem Scenario

- Thursday, 5:40 PM. Kim Subin is waiting for the company shuttle home when he remembers the external client meeting scheduled for the next morning. He usually wears a black hoodie and jeans to work, and it's never really been an issue at the office. But as face-to-face meetings with clients have become more frequent lately, he's started worrying more about what to wear. He still can't shake something his team lead said with a smile after the last meeting: "Subin, maybe throw on a shirt next time."
- Subin opens Musinsa and searches "men's client meeting outfit" and "men's business casual." But most of the results — shirts, slacks, blazers — are styles he almost never wears. Ever since he bought a black set and a navy blazer last month for a wedding, the recommendation algorithm has kept showing him formal clothing as well. Looking at the recommended items, he feels like they've drifted away from his usual taste.
- At the same time, showing up in his usual hoodie feels risky given what his team lead said and the impression he wants to make on the client. On the other hand, wearing the blazer feels overly formal like something that would make his junior coworkers ask, "Hyung, what's going on with you today?" What Subin needs isn't simply "formal clothes," but a way to figure out how far to dial up the formality for today's situation while still keeping his usual style intact.
- But that's hard to put into an exact search query. As he goes back and forth comparing multiple shopping sites and outfit photos, the time left before the shuttle arrives keeps shrinking. In the end, unable to reach a clear decision, Subin ends up reaching for his familiar black clothes again.

B. Detailed Scenario

1. Identifying the base style and the current situation

Subin launches our system. The system has a base style profile stored from Su-bin's earlier input:

- Prefers dark colors
- Prefers a relaxed fit
- Prefers designs with minimal logos, patterns, or decoration
- Prefers a casual style
- Values comfortable wear

Rather than modifying the base profile right away, **the agent treats it as background information for the current conversation.**

AI asks: "What situation are you dressing for today?"

Subin types: "I have a client meeting tomorrow. I don't really want to wear my usual hoodie, but I don't want to dress like I'm in a suit either."

From this input, the system extracts the current context: client meeting, need for a higher level of formality than usual, and avoiding excessive formality. At the same time, it determines that the exact level of formality Subin wants is still uncertain.

2. Exploring uncertain preferences through images

The system determines that comparing actual outfits will capture Subin's intent better than an additional text-based question like "Please rate the level of formality you want from 1 to 5."

Two outfit options are shown on screen

Outfit A: Black knit + relaxed black pants + minimal sneakers

Outfit B: White shirt + navy blazer + slacks + loafers

Rather than simply asking "Which outfit do you like better?", the agent frames the question to explicitly include the current situation.

AI asks: "If you were heading to tomorrow's client meeting, which of these would you lean toward wearing?"

Subin selects Like for Outfit A.

However, since Outfit A and Outfit B differ in several respects such as color, fit, outerwear, and shoes the system doesn't simply interpret every feature of A as Subin's preference. Judging that there's significant uncertainty about the reason behind his choice, it presents a short follow-up question.

AI asks: "What's the main reason you preferred Outfit A?"

- It looks comfortable
- It's less formal
- I like the color
- I like the overall vibe

Subin selects "It's less formal."

The system reflects this not as a long-term baseline preference, but as a requirement specific to the current client-meeting session.

3. Confirming necessary information through dialogue

Next, the system determines that while the amount of travel and physical activity involved matters for the current situation, this is information better gathered by asking directly rather than by showing images.

AI asks: "Will you be doing a lot of moving around or walking tomorrow?"

Subin replies: "I'll go from Pangyo to Gangnam and then head back to the office after the meeting. I think I'll be walking quite a bit."

The system adds activity level, extended wear, and maintaining comfort to the current context. It then summarizes what it has gathered so far and checks it with Subin.

AI says: "It sounds like you want to keep your usual comfortable, dark-toned style, but look a notch more polished than a hoodie **without feeling like you're in a suit.**"

Subin replies: "That's right."

4. Fine-tuning through visual comparison

The system has now sufficiently identified the key situational factors and Subin's baseline preferences, but it's still uncertain about exactly which way of raising the formality level feels most natural to him.

This time, it presents two outfits that are kept as similar as possible across other attributes, varying only the formality of the top.

Outfit A Charcoal crewneck knit + black pants + gray sneakers

Outfit B Dark gray overshirt + black pants + gray sneakers

Subin selects Outfit B.

From this, the system infers that Subin isn't rejecting a neater style itself. Rather, he finds formality uncomfortable when it breaks sharply from his usual style, as with a blazer or suit.

The agent checks this interpretation again.

AI says: "It seems like you'd rather raise the level of neatness in a way that flows naturally from your usual style like an overshirt rather than something like a blazer."

Subin confirms this.

5. Final recommendation and rationale

Based on Subin's baseline preferences and the situational requirements identified so far, the system proposes three outfit options.

The first outfit keeps the most elements of Subin's usual style; the second strikes a balance between his usual style and the demands of the meeting; and the third stays reasonably close to his existing taste while allowing him to try something a bit newer.

Subin makes his final choice: the second outfit.

Along with the recommendation, **the system distinguishes which elements were preserved from his existing preferences and which were adjusted for today's situation.**

Elements preserved from his usual preferences

- Dark colors
- Relaxed silhouette
- Design with minimal logos and decoration
- Comfortable sneakers

Elements adjusted for the current situation

- An overshirt in place of a hoodie
- A higher level of formality than usual
- A neater silhouette suited to the client meeting

Elements clarified through this conversation

- Blazer-level formality feels like too much
- A semi-casual style that connects naturally with his usual look is acceptable

Subin judges that with this recommended outfit, he won't come across as overly casual to the client, nor will he look out of place or awkward to his coworkers.

3-1

Persona #2

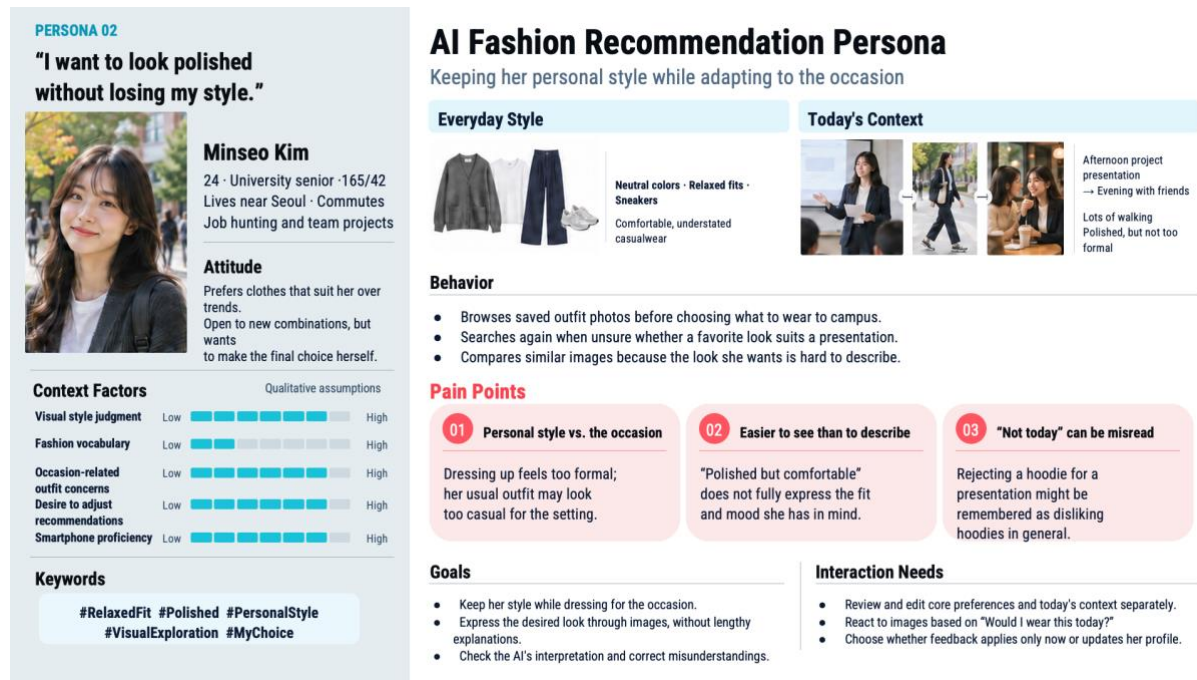


Figure2. Summary of Persona #2

Description

Minseo Kim is a 24-year-old university senior balancing job hunting and team projects. She prefers neutral colors, relaxed fits, and sneakers, valuing comfortable clothes that suit her rather than following trends. Before leaving for campus, she browses saved outfit photos but struggles to balance a polished appearance with her personal style on days that involve both a presentation and meeting friends. Although she can recognize the look she wants in images, describing it through fashion terms or detailed sentences feels difficult, so she repeatedly compares similar photos. She also does not want to reject an outfit for a presentation to be interpreted as a lasting dislike of that style. She therefore wants to adjust her everyday preferences and the day's context separately, refining her choices by reacting to outfit images. She expects to review and correct the AI's interpretation of her intentions and decide whether her feedback should apply only to the current recommendation or be saved as an ongoing preference.

3-2

Scenario #2

A. Problem scenario

- Monday afternoon, Minseo has a team project presentation at school the next day. Since the weather has turned cooler, she's made evening plans with friends in Seorae Village afterward browsing several small shops, taking photos, and walking around, so it won't be a short distance to cover. She usually likes a combination of a loose-fitting knit, wide pants, and sneakers, but for the presentation she doesn't want to look too relaxed or unpolished, since that might make her seem unprepared.
- Minseo first picks out a few outfit photos she's saved on Ably and tells Gemini "I have a project presentation tomorrow and I want to look neat without looking too much like I'm in a suit. I'm also meeting friends afterward." Gemini suggests a generic combination that looks presentation-appropriate — blazer, shirt, slacks. Minseo doesn't think it's a wrong answer exactly, but it feels far from the style she normally wears. She tries to explain more specifically that she wants to "look neat without losing my own style," but struggles to find the right words for it.

- She then searches Zigzag for "women's college presentation look" and "project presentation outfit." Filters and styled content bring up plenty of items, but the results are full of things like blouses, jackets, and loafers that seem tailored only to the presentation setting.
- On Ably, she also browses the "work outfit" and "semi-formal" categories, but she must keep manually balancing, on her own, the line between the comfortable casual items she's liked before and the neatness she needs today. Still, she can't keep spending this much time hunting for something she likes, since she also needs to do a final check of her presentation prep tomorrow so she ends up having to settle for a compromise. Situations like this have been happening more often lately, and it's been a real source of stress. Even though these are apps she uses all the time, she feels frustrated that there's no feature that can account for "my personal taste," "presentation formality," and "plans with friends" all at once.
- Minseo finds herself torn between the following:
 - Sticking with her usual style might make her look too casual for the presentation.
 - Choosing a jacket and loafers would feel unlike her and overdressed.
 - Factoring in the evening plans with friends makes it hard to decide based on the presentation alone.
 - She worries that telling Gemini "I don't want a blazer today" might be interpreted as disliking blazers altogether going forward.
- The next day, Minseo ends up wearing her familiar knit and an ordinary pair of pants, but she keeps second-guessing her outfit throughout the presentation. After somehow getting through it, she arrives at the meetup spot with her friends only to feel her mood sour a bit at the sight of her friends in stylish, trendy outfits that fit the Seorae Village vibe, making her own outfit feel a little underwhelming by comparison.
- On her way home, Minseo feels the need for an app that does more than just list products — one that could help her figure out, in situations like today's presentation, evening plans, and amount of walking involved, which elements to adjust and by how much while keeping her usual style intact.

B. Detailed scenario

1. User modeling layer: Confirming baseline preferences

Minseo launches the AI fashion recommendation agent. An editable baseline style profile appears first on screen.

- Prefers neutral/achromatic and muted colors
- Prefers a relaxed fit
- Prefers sneakers
- Prefers a comfortable but neat casual style
- Dislikes clothes with excessive logos/decoration
- Tends to judge by comparing images
- Wants to make the final choice herself

Rather than modifying the baseline profile right away, the **system treats it as background information for the current conversation.**

AI asks: "What situation are you dressing for today?"

Minseo types: "I have a project presentation at school tomorrow. My usual style feels too casual and unpolished for that... After the presentation, I have dinner plans with friends in Seorae Village — I'll have to walk a lot, and I want to keep my usual style while looking a bit neater."

The system extracts the following as the current session's context:

- Purpose: Team project presentation
- Following schedule: Dinner plans with friends
- Constraint: Extended travel and walking
- Desired impression: Neatness, looking prepared
- Impression to avoid: Usual overly casual look
- Baseline preferences to preserve: Comfortable fit, neutral colors, sneakers

For information that needs factual confirmation, like "having to walk a lot," the agent asks a short text-based question.

AI asks: "Will it be hard to change shoes after the presentation, and will you be on the move continuously?"

When Minseo replies, "Yeah, I want to wear sneakers all day"

the system reflects activity level and sneakers as today's essential constraints.

On the other hand, it judges that "neat, but still like me" is difficult to interpret from language alone.

2. Searching for uncertain preferences through images

Here, the agent opts for image-based comparison instead of another text question.

Outfit A: Charcoal knit + wide slacks + white sneakers

Outfit B: Black blazer + shirt + slacks + loafers

AI asks: "Considering both tomorrow's presentation and the dinner plans, which look would you lean toward?"

Minseo selects A.

Since it took her a bit longer to decide, the system doesn't assume she likes every feature of A, and instead presents a short follow-up question.

AI asks: "What's the main reason you picked A?"

→ It looks comfortable / It's less casual / I like the color / I like the overall vibe

Minseo selects "It looks comfortable" and "It's less casual."

At this point, the agent checks its interpretation.

AI says: "It sounds like you want to keep your usual comfortable silhouette and sneakers but tidy up the material and fit the knit or pants a bit more to suit the presentation. Is that right?"

Once Minseo confirms this, the system judges that further questions would yield diminishing informational value and moves on to the recommendation stage. **Here, it reflects this not as a long-term baseline preference, but as a requirement specific to the current presentation session.**

3. Confirming necessary information through dialogue

Based on the preceding context and choices, the system judges that image selection alone has limits for capturing the shifting context, and that it needs to ask an additional direct question.

AI asks: "Given the dinner plans, would you like to avoid anything too formal?"

Minseo replies: "After the presentation I have dinner plans with friends, and we'll be going to cafes, taking photos, and browsing shops in Seorae Village, so I don't want anything too suit-like."

The system adds casualness and activity level to the current context.

It then confirms what it has gathered so far with Minseo.

AI says: "It sounds like you want to keep your usual casual style but lean it toward the more formal end within that range."

Minseo selects "That's right."

4. Fine-tuning through visual comparison

The system has now identified the key situational factors and Minseo's baseline preferences, but it's still uncertain about exactly what degree of neatness and freshness she wants, given the balance between the presentation and the dinner plans.

The agent judges that image comparison will capture Minseo's intent better than another text question. So, it presents two outfits that keep all other elements as similar as possible, varying only the level of formality and mood.

Outfit A: Dark gray knit + wide black pants + white sneakers

Outfit B: Minimal overshirt + ivory inner T-shirt + wide slacks + black sneakers

AI asks: "Considering both the presentation and the dinner plans, which one feels closer to the look you want?"

Minseo selects Outfit B.

From this, the system infers that Minseo isn't uncomfortable with neatness itself, but rather with formality that breaks sharply from her usual style like a blazer or loafers. On the other hand, it judges that she can accept neatness achieved in a way that flows naturally from a comfortable, casual style, as with an overshirt.

The agent checks its interpretation again: "It seems like you'd prefer to keep your usual comfortable silhouette and sneakers, while raising the neatness in a natural way like with an overshirt. Is that right?" Minseo selects "That's right."

5. Final recommendation and rationale

Based on Minseo's baseline preferences and the situational requirements identified so far, the system presents three outfit options using product catalogs from Zigzag and Ably and pre-built outfit combinations.

The first outfit preserves the most elements of Minseo's usual style; the second balances her everyday preferences with the formality required for the presentation and her plans afterward; and the third stays within her preferred neutral palette while introducing a slightly more distinctive style.

Minseo makes her final choice: the second outfit.

Along with the recommendation, **the system distinguishes which elements were preserved from her existing preferences and which were adjusted for today's situation.**

Elements preserved from her usual preferences

- Relaxed silhouette
- Neutral and low-saturation colors
- Minimal logos and decorative elements
- Comfortable sneakers

Elements adjusted for the current situation

- A neater upper-body silhouette for the presentation
- A slightly higher level of formality than usual
- Styling that can transition naturally from the presentation to dinner plans

Elements clarified through this conversation

- Blazer-level formality feels unnecessarily stiff
- She prefers to maintain a relaxed silhouette even when dressing more formally
- Muted accent colors are acceptable when they do not make the overall outfit feel too conspicuous

Minseo judges that the second outfit allows her to look sufficiently prepared and professional during the presentation while still feeling like herself. She also feels that the outfit is comfortable and casual enough to continue wearing during the dinner plans afterward.

4

Claim Analysis

Design feature	Upside (+)	Downside / risk (-)
Editable base style profile used only as background supports H1	Preserves the user's stable taste and sense of control; removes the need to re-explain taste each session and eases cold-start.	If the profile is outdated it biases recommendations; reviewing and maintaining it becomes an upfront effort.
Session-scoped memory with an explicit "remember this?" choice supports H1	Prevents a temporary need from contaminating overall taste (solves the formal-recommendation flood after a one-off wedding purchase); letting the user control the learning scope builds trust.	Strict per-session isolation may never learn a genuinely recurring preference; deciding "keep or not" each time adds cognitive load.
Adaptive switching between dialogue and outfit-image cards supports H2	Captures hard-to-verbalize taste (formality, mood) through low-effort visual comparison, so the user feels understood; captures factual constraints (mobility) quickly through text.	A wrong modality choice (asking in text what should be shown) increases frustration; frequent switching can feel scattered.
"Biggest reason" follow-up after a like/dislike selection supports H2	Resolves the multi-attribute confound of an image choice, avoiding wrong generalization and attributing the preference correctly; verbalizing the reason aids self-understanding.	Adds question fatigue; if the true reason is not among the offered options, the response is distorted.